Home Work 2
II B. Tech. (Common)
Mathematics-II (MCI102)

Notations:

$M_{m\times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.
$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.
$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.
$C(\mathbb{R})$: The set of all real-valued continous functions
$\mathbb{R}^\infty$: The set of all real-valued sequences.
$\mathbb{R}^+$: The set of positive real numbers.

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1. Which of the following are vector spaces?

   (a) $V = \mathbb{R}^2$ over $\mathbb{R}$ with componentwise addition and $\lambda \odot (x, y) = (3\lambda x, y)$ for all $(x, y) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$.

   (b) $V = \mathbb{R}^+$ over $\mathbb{R}$ with $x \oplus y = xy$ and $\lambda \odot x = x^\lambda$ for all $x, y \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$.

   (c) $V = \mathbb{R}[x]$ over $\mathbb{R}$ with usual addition and scalar multiplication of polynomials.

   (d) $V = \mathbb{R}^\infty$ over $\mathbb{R}$ with $a + b = \{a_n + b_n\}_{n=1}^\infty$ and $\lambda a = \{\lambda a_n\}_{n=1}^\infty$ for all $a = \{a_n\}_{n=1}^\infty$, $b = \{b_n\}_{n=1}^\infty$ $\in \mathbb{R}^\infty$ and $\lambda \in \mathbb{R}$.

   (e) $V = \{t_\alpha : \mathbb{R} \to \mathbb{R} \mid t_\alpha(x) = x + \alpha, \alpha \in \mathbb{R}\}$ over $\mathbb{R}$ with composition of mappings and $\lambda t_\alpha = t_{\alpha\lambda}$ for all $t_\alpha \in V$ and $\lambda \in \mathbb{R}$.

   (f) $V = C[a, b]$ over $\mathbb{R}$ with $(f + g)(x) = f(x) + g(x)$ and $(\lambda \cdot f)(x) = \lambda f(x)$ for all $\lambda \in \mathbb{R}, f, g \in C[a, b]$.

   (g) V is the set of all real-valued continous functions defined on an open interval I which have at most finite number of points of discontinuity over $\mathbb{R}$ with pointwise addition and scalar multiplication of functions.

   (h) $V = \{$ all real polynomials of degree 4 or 6$\}$ over $\mathbb{R}$ with usual addition and scalar multiplication.

   (i) $V = \{$ all $n \times n$ Hermitian matrices$\}$ over $\mathbb{C}$ with usual addition and scalar multiplication.

   (j) $V = \{$all $n \times n$ skew-Hermitian matrices$\}$ over $\mathbb{C}$ with usual matrix addition and scalar multiplication.

   Answers: (a) No (b) Yes (c) Yes (d) Yes (e) Yes (f) Yes (g) Yes (h) No (i) No (j) No.

2. On $\mathbb{R}^2$, define two operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 - x_2, y_1 - y_2),$$

$$c \odot (x, y) = (-cx, -cy)$$

   where $c \in \mathbb{R}$. Which of the axioms for a vector space are satisfied by $(\mathbb{R}^2, \oplus, \odot)$ over $\mathbb{R}$?

3. On $\mathbb{R}^2$, define two operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, 0),$$

$$c \odot (x, y) = (cx, 0)$$

   where $c \in \mathbb{R}$. Is $\mathbb{R}^2$, with these operations, a vector space over $\mathbb{R}$?